

MATH3303 – Lecture 12

Classification of Finite Abelian Groups III: Decomposition into p -Groups

Review Notes

Reference: Lee, Abstract Algebra, §5.2

Overview

This lecture continues building towards the Fundamental Theorem of Finite Abelian Groups:

- **Completion of Lemma 5** (Lee, Lemma 5.4): verifying that the second factor $(a^{p^m})^u$ in the decomposition has order dividing q (fewer prime divisors), completing the inductive step.
- **Lemma 6** (Lee, Lemma 5.5): every finite abelian group decomposes as a direct product of its p_i -subgroups, one for each prime p_i dividing $|G|$.
- **Lemma 7** (Lee, Lemma 5.6): the key technical lemma — in a finite abelian p -group, an element of largest order splits off as a direct factor: $G = \langle g \rangle \times H$.

1 Completion of Lemma 5

Recall from Lecture 11 (Lemma 4, Lee Lemma 5.4): for a non-identity element a of finite order in any group G , writing $|a| = p^m q$ with $\gcd(p, q) = 1$ and choosing $u, v \in \mathbb{Z}$ with $p^m u + qv = 1$:

$$a = (a^q)^v \cdot (a^{p^m})^u.$$

We showed $(a^q)^v$ is a p -element. We now verify $(a^{p^m})^u$ has the right form for the inductive hypothesis to apply.

Completion of Lemma 5 proof. $(a^{p^m})^u$ **has order dividing q .**

Compute: $(a^{p^m})^q = a^{p^m q} = a^{|a|} = 1$. Hence $|a^{p^m}|$ divides q .

Since q has *fewer distinct prime divisors* than $|a| = p^m q$ (as p no longer appears), the inductive hypothesis applies to a^{p^m} : it can be expressed as a product of p_i -elements for primes p_i dividing q .

Therefore $(a^{p^m})^u$ is also such a product, and combining:

$$a = (a^q)^v \cdot (a^{p^m})^u$$

is a product of p_i -elements for primes p_i dividing $|a|$. □

2 Lemma 6: Decomposition into p -Subgroups

Lemma 2.1: Finite abelian group is a direct product of p -subgroups (Lemma 6 from lecture; cf. Lee, Lemma 5.5)

Let G be a nontrivial finite abelian group, and let $p_1, p_2, \dots, p_k$ be the distinct primes dividing $|G|$. Then

$$G \cong H_1 \times H_2 \times \cdots \times H_k,$$

where each H_i is the subgroup of G consisting of all p_i -elements, and each H_i is a p_i -group.

Proof. By Lemma 3 (Lecture 11, Lee Lemma 5.3), each H_i is a subgroup of G . Since G is abelian, every subgroup is normal, so each $H_i \trianglelefteq G$.

We verify the three conditions for G to be the internal direct product of $H_1, \dots, H_k$ (then apply Lemma 2 / Lee Thm. 5.2).

Condition 1: each $H_i \trianglelefteq G$. Done above.

Condition 2: $G = H_1 H_2 \cdots H_k$.

For any $g \in G$: since g has finite order dividing $|G|$, by Lemma 5 (Lecture 11, Lee Lemma 5.4), g is a product of p_i -elements for primes p_i dividing $|g|$ (hence dividing $|G|$). Each such p_i -element lies in H_i . So $g \in H_1 H_2 \cdots H_k$.

(If $g = 1$, there is nothing to check since $1 \in H_i$ for each i .)

Condition 3: $(H_1 H_2 \cdots H_i) \cap H_{i+1} = \{1\}$ for $1 \leq i < k$.

Suppose $g \in H_{i+1}$ and simultaneously $g = g_1 g_2 \cdots g_i$ with $g_j \in H_j$.

Let $|g_j| = p_j^{m_j}$ and set

$$m := p_1^{m_1} p_2^{m_2} \cdots p_i^{m_i}.$$

Then $g^m = g_1^m g_2^m \cdots g_i^m = 1$, since each $|g_j| = p_j^{m_j}$ divides m . (We use abelian here to write $(g_1 \cdots g_i)^m = g_1^m \cdots g_i^m$.)

Thus $|g|$ divides m . But $g \in H_{i+1}$ is a p_{i+1} -element, so $|g| = p_{i+1}^n$ for some $n \geq 0$.

Since $\gcd(m, p_{i+1}) = 1$ (the primes $p_1, \dots, p_k$ are distinct), the only way $p_{i+1}^n \mid m$ is if $n = 0$, i.e. $|g| = 1$, i.e. $g = 1$.

Hence $(H_1 \cdots H_i) \cap H_{i+1} = \{1\}$.

Conclusion. G is the internal direct product of $H_1, \dots, H_k$, and by Lemma 2 (Lee Thm. 5.2):

$$G \cong H_1 \times H_2 \times \cdots \times H_k. \quad \square$$

Remark 2.1: What Lemma 6 achieves (from lecture)

Lemma 6 reduces the classification of finite abelian groups to classifying finite abelian p -groups for each prime p . This is exactly what Lemma 7 and the Fundamental Theorem address next.

3 Lemma 7: Splitting off the Largest-Order Element

The following is the most technical lemma in the classification proof. It shows that in any finite abelian p -group, an element of largest order splits off as a direct cyclic factor.

Lemma 3.1: Splitting off largest-order element (Lemma 7 from lecture; cf. Lee, Lemma 5.6)

Let G be a finite abelian p -group, and let $g \in G$ be an element of largest possible order. Then

$$G \cong \langle g \rangle \times H \quad \text{for some subgroup } H \leq G.$$

Proof. By strong induction on $|G|$.

Base case $|G| = 1$: Then $g = 1$ and $H = \{1\}$ works.

Inductive step: Assume $|G| > 1$ and the lemma holds for all abelian p -groups of smaller order. Let $|g| = p^n$ with $n \geq 1$.

Trivial case $\langle g \rangle = G$: Use $H = \{1\}$.

Nontrivial case $\langle g \rangle \subsetneq G$: Take $b \in G \setminus \langle g \rangle$. Since b is a p -element, $b^{p^k} = 1 \in \langle g \rangle$ for some $k \geq 1$.

Let m be the *smallest* positive integer such that $b^{p^m} \in \langle g \rangle$, and let $c := b^{p^{m-1}}$. Then:

- $c \notin \langle g \rangle$ (by minimality of m);
- $c^p = b^{p^m} \in \langle g \rangle$, so write $c^p = g^i$ for some $i \in \mathbb{Z}$.

Step 1: $p \mid i$.

Since G is a p -group with largest order p^n , $c^{p^n} = 1$, so $|c^p|$ divides p^{n-1} .

Suppose for contradiction $\gcd(p, i) = 1$. Then by Lee Corollary 3.2, $|c^p| = |g^i| = |g| = p^n$, which contradicts $|c^p| \leq p^{n-1}$. Hence $p \mid i$; write $i = pj$.

Step 2: Construct $d := g^{-j}c$.

Since $g^j \in \langle g \rangle$: if $d \in \langle g \rangle$, then $c = g^j d \in \langle g \rangle$, contradicting $c \notin \langle g \rangle$. So $d \notin \langle g \rangle$.

Compute: $d^p = g^{-jp}c^p = (g^i)^{-1} \cdot g^i = 1$. Hence $|d| = p$.

Step 3: Pass to $M = G/\langle d \rangle$.

Since G is abelian, $\langle d \rangle \trianglelefteq G$ (and $|d| = p$ so $|\langle d \rangle| = p$). The quotient $M = G/\langle d \rangle$ is:

- abelian (quotient of abelian is abelian);
- a p -group (orders of elements divide those in G , hence are powers of p);
- of order $|G|/p < |G|$.

Step 4: $g\langle d \rangle$ has order p^n in M .

Suppose $(g\langle d \rangle)^{p^{n-1}} = \langle d \rangle$, i.e. $g^{p^{n-1}} \in \langle d \rangle$. Since $g^{p^{n-1}} \neq 1$ (as $|g| = p^n$), write $g^{p^{n-1}} = d^s$ with $0 < s < p$.

Since $\gcd(s, p) = 1$, Bezout gives $u, v \in \mathbb{Z}$ with $su + pv = 1$, so $d = d^{su+pv} = (d^s)^u (d^p)^v = g^{p^{n-1}u} \in \langle g \rangle$. But $d \notin \langle g \rangle$ — contradiction. Hence $|g\langle d \rangle| = p^n$.

Step 5: Apply inductive hypothesis to M .

Since $|M| < |G|$ and $g\langle d \rangle$ has largest possible order p^n in M , by the inductive hypothesis:

$$M = \langle g\langle d \rangle \rangle \times K \quad \text{for some } K \leq M.$$

By the Correspondence Theorem (Lee, Thm. 4.8), $K = H/\langle d \rangle$ for some subgroup $H \leq G$ containing $\langle d \rangle$.

Step 6: Show $G = \langle g \rangle \times H$.

Trivial intersection $\langle g \rangle \cap H = \{1\}$: If $g^i \in \langle g \rangle \cap H$, then $g^i \langle d \rangle \in \langle g \langle d \rangle \rangle \cap K = \{e_M\}$, so $g^i \in \langle d \rangle$. Since $\langle d \rangle$ has order p and $|\langle g \rangle| = p^n$: $g^i \in \langle d \rangle \cap \langle g \rangle$. If $g^i \neq 1$, then $|g \langle d \rangle|$ would be reduced below p^n (contradiction). So $g^i = 1$.

$G = \langle g \rangle H$: Take any $x \in G$. In $M = \langle g \langle d \rangle \rangle \times K$, write $x \langle d \rangle = g^t \langle d \rangle \cdot w \langle d \rangle$ for some $t \in \mathbb{Z}$, $w \in H$. Then $x = g^t w d^\ell$ for some ℓ . Since $d \in \langle d \rangle \subseteq H$, we have $d^\ell \in H$, so $w d^\ell \in H$. Hence $x \in \langle g \rangle H$.

Therefore $G = \langle g \rangle \times H$. □

Remark 3.1: Why the proof is subtle (from lecture)

The main challenge is that we cannot directly split G into $\langle g \rangle$ and some complement — the complement might not exist in G directly. Instead, the proof:

- (1) Finds an element d of order p outside $\langle g \rangle$, with $d^p = 1$.
- (2) Passes to the smaller group $M = G/\langle d \rangle$, where induction gives a splitting.
- (3) Lifts the splitting back to G using the Correspondence Theorem.

The key constraint used throughout is that g has *largest* order in G — this is what forces $p \mid i$ in Step 1 and preserves the order of $g \langle d \rangle$ in M in Step 4.

Checklist of Skills

- ☐ Complete the proof of Lemma 5 (Lee, Lemma 5.4): verify $|a^{p^m}|$ divides q and that q has fewer prime divisors than $|a|$, allowing induction.
- ☐ State and prove Lemma 6 (Lee, Lemma 5.5): a finite abelian group is the direct product of its p_i -subgroups. Verify all three internal direct product conditions.
- ☐ In the proof of Condition 3 of Lemma 6: set up $m = p_1^{m_1} \cdots p_i^{m_i}$, use abelian to compute $g^m = 1$, then use coprimality to conclude $g = 1$.
- ☐ Explain why Lemma 6 reduces the classification to p -groups.
- ☐ State Lemma 7 (Lee, Lemma 5.6): in a finite abelian p -group, $G = \langle g \rangle \times H$ for the element g of largest order.
- ☐ Outline the strategy of the proof: find d of order p outside $\langle g \rangle$, pass to $M = G/\langle d \rangle$, apply induction, lift via Correspondence Theorem.
- ☐ Explain the role of Step 1 ($p \mid i$): it allows construction of d with $d^p = 1$.
- ☐ Explain why the g having largest order is needed: it ensures $|g\langle d \rangle| = p^n$ is preserved in the quotient M .